

A PROJECT REPORT
ON
DATA VISUALIZATION WITH PYTHON



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INTRODUCTION

Data visualization is the graphical representation of data using charts, graphs, maps, and dashboards to make information easier to understand and analyze. It helps in identifying patterns, trends, relationships, and insights from large datasets in a simple and interactive manner. In today's data-driven world, data visualization plays a crucial role in decision-making across various fields such as business, healthcare, transportation, finance, and research. By converting raw data into visual formats, complex information becomes more meaningful and accessible to users.

Python is one of the most widely used programming languages for data analysis and visualization because of its simplicity, flexibility, and powerful libraries. Libraries such as **Pandas** help in data cleaning and analysis, while visualization libraries like **Matplotlib**, **Seaborn**, and **Plotly** allow users to create interactive and visually appealing charts and dashboards. Python also supports integration with tools like **Jupyter Notebook**, **Power BI**, and web frameworks, making it highly suitable for analytical projects. Its open-source nature and strong community support further contribute to its popularity among data analysts and researchers.

This project focuses on the **Road Accident Fatality Analysis Dashboard** using Python for data visualization. The dashboard analyzes state-wise road accident fatality data in India from 2020 to 2024. The objective of the project is to identify accident trends, compare fatalities across different states, and provide meaningful insights into road safety conditions in the country. Using Python visualization libraries, interactive charts and graphs are created to present the data in an effective manner. The dashboard helps users easily understand which states have the highest and lowest accident fatalities, observe yearly changes, and support awareness and decision-making related to road safety and traffic management.

OBJECTIVES OF THE PROJECT

The primary objective of this project is to develop a **Road Accident Fatality Analysis Dashboard** using Python and data visualization techniques. The project aims to analyze state-wise road accident fatality data in India from 2020 to 2024 and present it in an interactive and easy-to-understand visual format.

1. **To analyze state-wise road accident fatality data from 2020 to 2024.**
The project studies accident fatalities across different states and union territories in India. This helps in understanding the overall trend of road accidents over the years.
2. **To visualize accident trends using charts and dashboards.**
The project converts raw data into visual formats such as graphs and charts. This makes the information easier to understand and compare.
3. **To identify states with high and low accident fatalities.**
The dashboard helps in comparing accident fatality rates among states. This can support better road safety planning and awareness.
4. **To compare yearly changes in accident fatalities.**
The project examines the increase or decrease in fatalities each year. This helps in identifying important trends and patterns.
5. **To present complex data in a simple and interactive format.**
Large datasets are simplified using dashboards and visual tools. This improves clarity and user understanding.
6. **To use Python libraries for data analysis and visualization.**
The project uses libraries such as Pandas, Matplotlib, Seaborn, and Plotly. These tools help in efficient data handling and visualization.
7. **To identify patterns that may help improve road safety measures.**
The analysis helps in recognizing high-risk regions and accident trends. These insights may help in improving traffic safety strategies.
8. **To demonstrate the practical use of data visualization in real-world analysis.**
The project shows how visualization techniques can be applied in transportation and safety studies. It also highlights the importance of data analytics in decision-making.

TOOLS & TECHNIQUES USED

The following tools and techniques were used in the development of the **Road Accident Fatality Analysis Dashboard** project:

1. **Python**
Python was used as the main programming language for data analysis and visualization. It provides simple syntax and powerful libraries for handling large datasets efficiently.
2. **Pandas**
Pandas was used for data cleaning, organizing, and analyzing the road accident dataset. It helped in managing tabular data and performing calculations easily.
3. **Matplotlib**
Matplotlib was used to create basic charts and graphs for visualizing accident fatality trends. It helped in presenting data in a clear graphical format.
4. **Seaborn**
Seaborn was used for creating advanced and visually attractive statistical graphs. It improved the overall appearance and readability of the visualizations.
5. **Plotly**
Plotly was used to create interactive charts and dashboards. It allowed better user interaction and dynamic visualization of accident data.
6. **Jupyter Notebook**
Jupyter Notebook was used as the development environment for writing and executing Python code. It helped in combining code, outputs, and visualizations in a single workspace.
7. **Data Visualization Techniques**
Techniques such as bar charts, line graphs, pie charts, and comparative analysis were used to represent the accident data visually. These techniques made the information easier to interpret.

DATASET DESCRIPTION

The dataset used in this project is a **Road Accident Fatalities dataset** containing state-wise information about deaths caused due to road accidents in India from the years **2020 to 2024**. The dataset is structured in tabular form and is suitable for data analysis and visualization tasks.

In this project, the dataset is utilized to perform **road safety analysis** using Python libraries and data visualization techniques. The dataset helps in understanding accident trends, comparing fatalities across states and union territories, and identifying regions with high accident rates. The cleaned and well-organized dataset can further be used for research, policy analysis, and transportation safety studies

```
import pandas as pd

# Load the dataset
df = pd.read_excel('Road Accident fatality dataset.xlsx')

# Display first five rows
df.head()
```

	S.No	State/UT	2020	2021	2022	2023	2024
0	1.0	Andhra Pradesh	7039.0	8186	8293	8137	8346
1	2.0	Arunachal Pradesh	73.0	157	148	145	168
2	3.0	Assam	2629.0	3036	2994	3296	3351
3	4.0	Bihar	6699.0	7660	8898	8873	9347
4	5.0	Chhattisgarh	4606.0	5371	5834	6166	6945

The above dataset provides an overview of road accident fatalities recorded in different states and union territories of India over a five-year period.

The following are the columns present in the dataset:

1. **S.No.** – Represents the serial number assigned to each state or union territory.
2. **States/UTs** – Contains the names of Indian states and union territories included in the dataset.
3. **2020** – Shows the total number of road accident fatalities recorded in the year 2020.
4. **2021** – Represents the number of fatalities reported in the year 2021.
5. **2022** – Displays the road accident fatality data for the year 2022.
6. **2023** – Contains the total fatalities recorded during the year 2023.
7. **2024** – Shows the latest available road accident fatality data for the year 2024.
8. **Total Fatalities** – Represents the combined number of fatalities across all years for each state/UT (can be calculated for analysis purposes).
9. **Yearly Trend Analysis** – Used to identify the increase or decrease in accident fatalities over different years.
10. **State-wise Comparison** – Helps compare accident fatality rates among different states and union territories.

All these features are numerical or categorical in nature and are suitable for statistical analysis, trend identification, and dashboard visualization using Python tools and libraries.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is an important step used in this project to understand and analyze the road accident fatality dataset before creating visualizations and dashboards. EDA helps in examining the structure, quality, and patterns present in the data. It allows the user to gain meaningful insights from raw data and prepares the dataset for effective visualization and analysis.

In this project, EDA was performed on the state-wise road accident fatality data of India from 2020 to 2024. The process included checking the dataset for missing values, understanding the distribution of accident fatalities, comparing yearly changes, and analyzing state-wise variations in fatalities. Through EDA, it became easier to identify trends such as states with consistently high accident fatalities, regions with increasing or decreasing fatality rates, and overall growth in road accident deaths over the years.

Various Python libraries such as Pandas, Matplotlib, and Seaborn were used to perform EDA tasks efficiently. Techniques like summary statistics, data grouping, trend analysis, correlation checking, and comparative analysis were applied to explore the dataset in detail. Visual methods such as bar charts, line graphs, and heatmaps were also used during the exploratory phase to better understand the data patterns.

EDA played a significant role in improving the quality and accuracy of the dashboard by helping in data cleaning, organization, and interpretation. The insights obtained during exploratory analysis supported the creation of meaningful visualizations and made the final dashboard more informative, interactive, and useful for understanding road safety trends in India.

IMPORTING REQUIRED LIBRARIES

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Set style for plots
plt.style.use('ggplot')
sns.set_theme(style='whitegrid')
```

The above code imports the required Python libraries used for data analysis and visualization in this project. **Pandas** and **NumPy** are used for handling and processing the dataset, while **Matplotlib**, **Seaborn**, and **Plotly** are used to create graphical visualizations and interactive dashboards.

READING THE DATASET

```
import pandas as pd

# Load the dataset
df = pd.read_excel('Road Accident fatality dataset.xlsx')

# Display first five rows
df.head()
```

	S.No	State/UT	2020	2021	2022	2023	2024
0	1.0	Andhra Pradesh	7039.0	8186	8293	8137	8346
1	2.0	Arunachal Pradesh	73.0	157	148	145	168
2	3.0	Assam	2629.0	3036	2994	3296	3351
3	4.0	Bihar	6699.0	7660	8898	8873	9347
4	5.0	Chhattisgarh	4606.0	5371	5834	6166	6945

The dataset is loaded into a **Pandas DataFrame** using the `read_excel()` function. Displaying the first few rows helps in understanding the structure of the dataset and the available features such as state names and yearly road accident fatality records from 2020 to 2024.

DATASET INFORMATION

```
▶ # Dataset information
print("Data Info:")
df.info()

# Descriptive statistics
print("\nDescriptive Statistics:")
df.describe()

# Checking missing values
print("\nMissing Values:")
df.isnull().sum()
```

```
Data Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   S.No         36 non-null    float64
1   State/UT     37 non-null    object
2   2020         36 non-null    float64
3   2021         37 non-null    int64
4   2022         37 non-null    int64
5   2023         37 non-null    int64
6   2024         37 non-null    int64
dtypes: float64(2), int64(4), object(1)
memory usage: 2.2+ KB
```

Descriptive Statistics:

Missing Values:

0

S.No	1
------	---

State/UT	0
----------	---

2020	1
------	---

2021	0
------	---

2022	0
------	---

2023	0
------	---

2024	0
------	---

dtype: int64

The dataset consists of **state-wise road accident fatality records in India** from 2020 to 2024. Most of the dataset is complete and properly structured, with only one missing value present in the 2020 column due to unavailable data for Ladakh. The dataset is clean and suitable for exploratory data analysis and visualization purposes.

STATISTICAL SUMMARY OF THE DATASET



```
# Statistical summary of the dataset
df.describe()
```

...	S.No	2020	2021	2022	2023	2024
count	36.000000	36.000000	37.000000	37.000000	37.000000	37.000000
mean	18.500000	7687.944444	8322.810811	9107.621622	9345.405405	9577.135135
std	10.535654	22898.749329	25148.970276	27528.077479	28256.154013	28954.838678
min	1.000000	0.000000	1.000000	2.000000	0.000000	0.000000
25%	9.750000	139.750000	140.000000	148.000000	145.000000	168.000000
50%	18.500000	2804.000000	3036.000000	2994.000000	3296.000000	3351.000000
75%	27.250000	6744.750000	7557.000000	7618.000000	7854.000000	7949.000000
max	36.000000	138383.000000	153972.000000	168491.000000	172890.000000	177177.000000

The statistical summary provides key numerical measures such as count, mean, standard deviation, minimum, maximum, and quartile values for the road accident fatality dataset. These statistics help in understanding the overall distribution, variation, and trends in accident fatalities across different states and years.

INSIGHTS & VISUALIZATION

INSIGHT 1: YEAR-WISE ROAD ACCIDENT FATALITIES

DISTRIBUTION

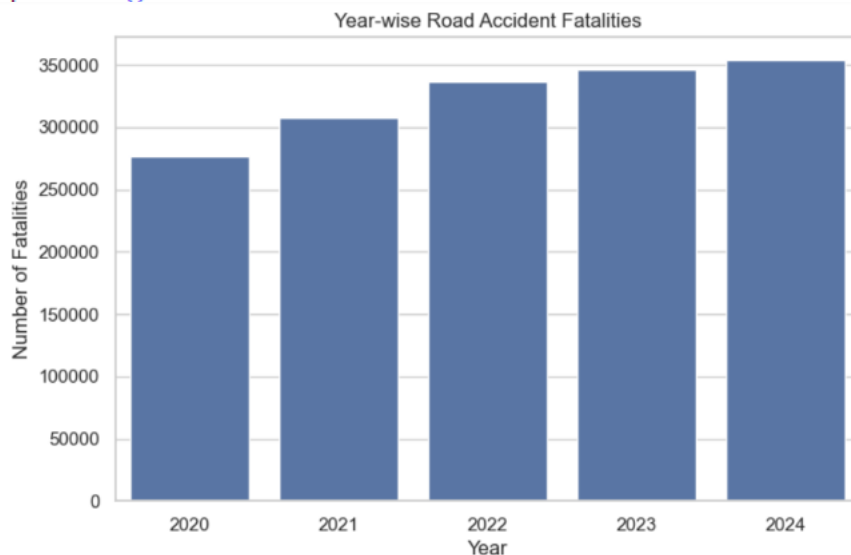
```
# Import required libraries
import matplotlib.pyplot as plt
import seaborn as sns

# Year-wise road accident fatalities distribution
years = ['2020', '2021', '2022', '2023', '2024']
totals = df[years].sum()

# Create visualization
plt.figure(figsize=(8,5))
sns.barplot(x=years, y=totals.values)

# Add labels and title
plt.title("Year-wise Road Accident Fatalities")
plt.xlabel("Year")
plt.ylabel("Number of Fatalities")

# Display graph
plt.show()
```



Insight-The above visualization shows the distribution of total road accident fatalities in India from 2020 to 2024. The graph highlights a continuous increase in accident fatalities over the years, indicating growing road safety concerns and the need for stronger traffic management and preventive measures across the country.

INSIGHT 2: STATE-WISE ROAD ACCIDENT FATALITIES (2024)

```
# Import required library
import matplotlib.pyplot as plt

# Remove total row
data = df[df['State/UT'] != 'Total (India)']

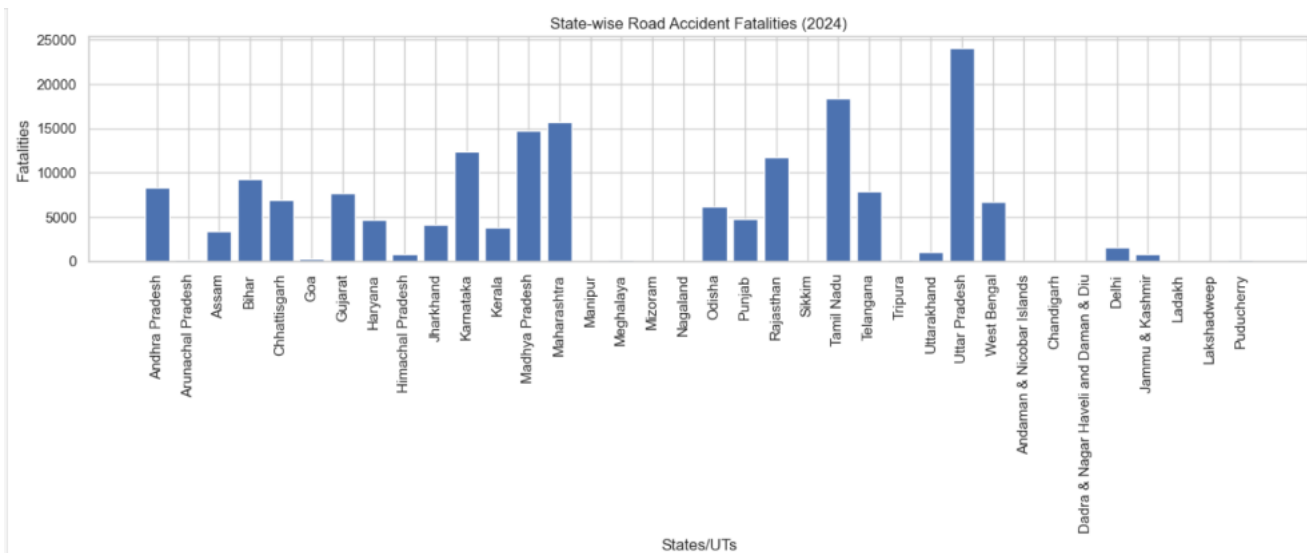
# Plot bar chart
plt.figure(figsize=(14,6))
plt.bar(data['State/UT'], data['2024'])

# Labels and title
plt.title("State-wise Road Accident Fatalities (2024)")
plt.xlabel("States/UTs")
plt.ylabel("Fatalities")

# Rotate labels
plt.xticks(rotation=90)

# Adjust layout
plt.tight_layout()

# Show graph
plt.show()
```



Insight-The bar chart compares road accident fatalities across different states and union territories in 2024. Some states recorded significantly higher fatalities, showing uneven road safety conditions across India.

INSIGHT 3: TOP 10 STATES WITH HIGHEST ROAD ACCIDENT FATALITIES (2024)

```
▶ # Import required libraries
import matplotlib.pyplot as plt

# Remove total row if present
data = df[df['State/UT'] != 'Total (India)']

# Select top 10 states with highest fatalities in 2024
top_states = data.sort_values(by='2024', ascending=False).head(10)

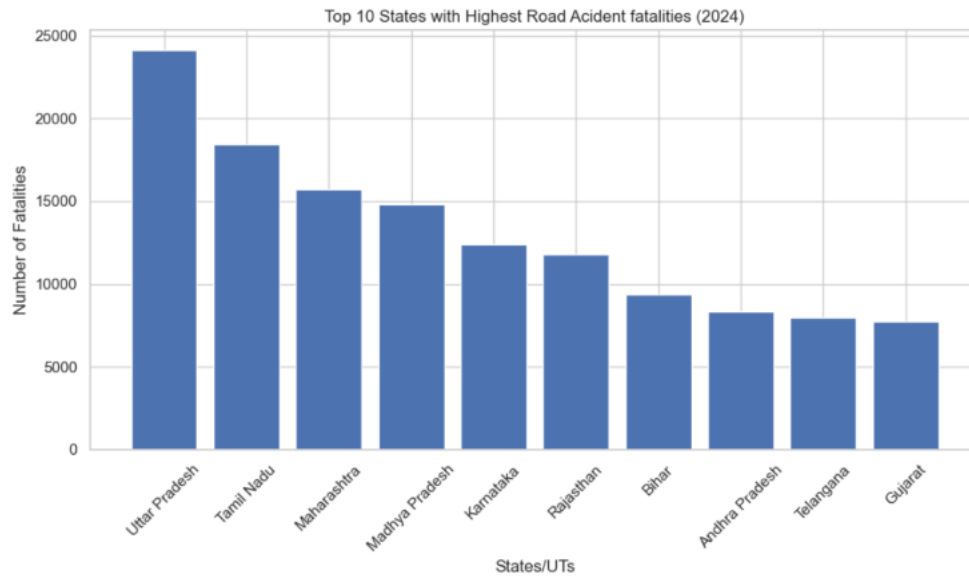
# Create bar chart
plt.figure(figsize=(10,6))
plt.bar(top_states['State/UT'], top_states['2024'])

# Add title and labels
plt.title("Top 10 States with Highest Road Accident Fatalities (2024)")
plt.xlabel("States/UTs")
plt.ylabel("Number of Fatalities")

# Rotate x-axis labels
plt.xticks(rotation=45)

# Adjust layout
plt.tight_layout()

# Display graph
plt.show()
```



Insight-The visualization highlights the states with the highest road accident fatalities in 2024. Uttar Pradesh, Tamil Nadu, and Maharashtra are among the most affected states with very high fatality numbers.

INSIGHT 4: TOP 10 STATES WITH LOWEST ROAD ACCIDENT FATALITIES (2024)

```
import matplotlib.pyplot as plt

# Top 10 lowest states
low_states = df.nsmallest(10, '2024')

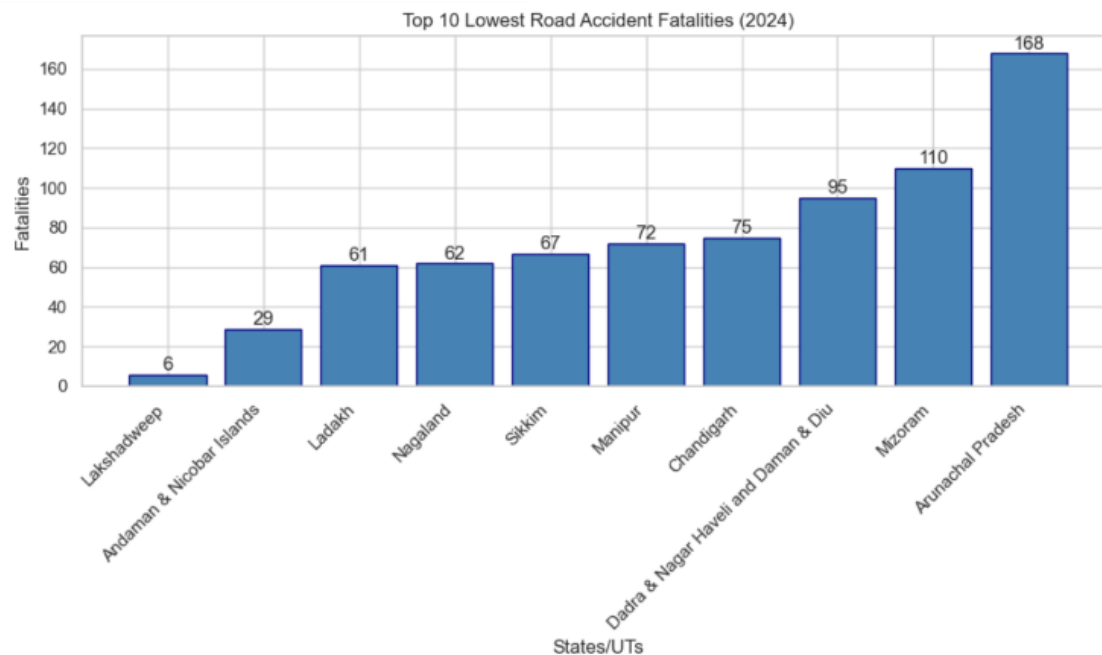
# Figure size
plt.figure(figsize=(10, 6))

# Bar chart
bars = plt.bar(
    low_states['State/UT'],
    low_states['2024'],
    color='steelblue',
    edgecolor='navy'
)

plt.title("Top 10 Lowest Road Accident Fatalities (2024)")
plt.xlabel("States/UTs")
plt.ylabel("Fatalities")
plt.xticks(rotation=45, ha='right')

# Values on top of bars
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 2,
        f'{int(height)}',
        ha='center'
    )

plt.tight_layout()
plt.show()
```



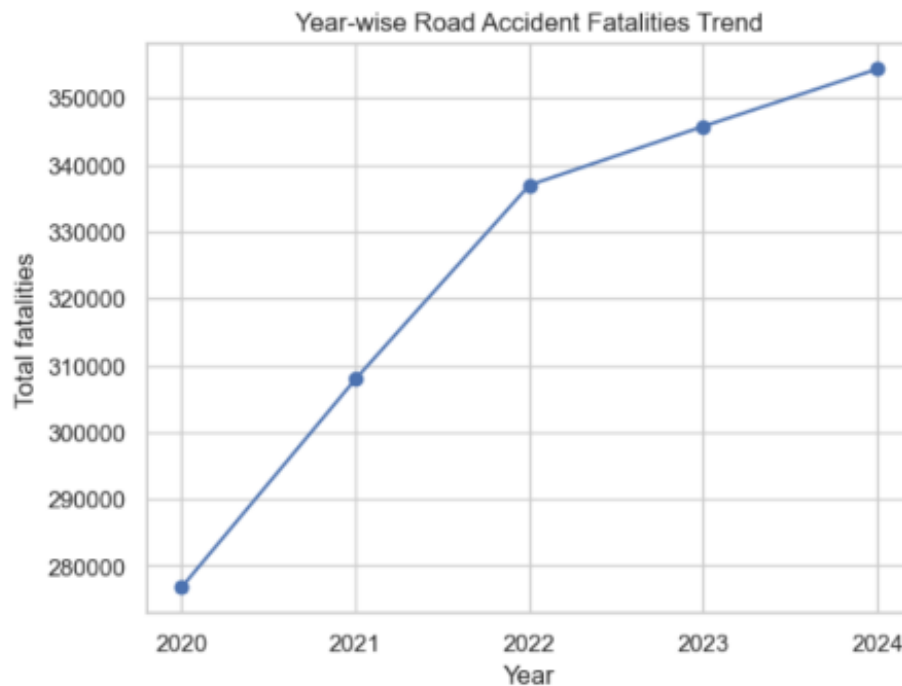
Insight-The bar chart shows the states and union territories with the lowest road accident fatalities in 2024. These regions recorded fewer accident deaths compared to other states, indicating relatively better road safety conditions or lower traffic volume.

INSIGHT 5 : AVERAGE ROAD ACCIDENT FATALITIES OF STATES (2020–2024)

```
import matplotlib.pyplot as plt

# Total fatalities year-wise
totals = df[['2020', '2021', '2022', '2023', '2024']].sum()

# Line chart
plt.plot(totals.index, totals.values, marker='o')
plt.title("Year-wise Road Accident Fatalities Trend")
plt.xlabel("Year")
plt.ylabel("Total Fatalities")
plt.show()
```



Insight-The graph shows a continuous rise in road accident fatalities from 2020 to 2024. The highest number of fatalities was recorded in 2024, indicating increasing road safety concerns. This trend highlights the need for stricter traffic rules, safer roads, and improved awareness programs.

INSIGHT 6: HEATMAP OF ROAD ACCIDENT FATALITIES

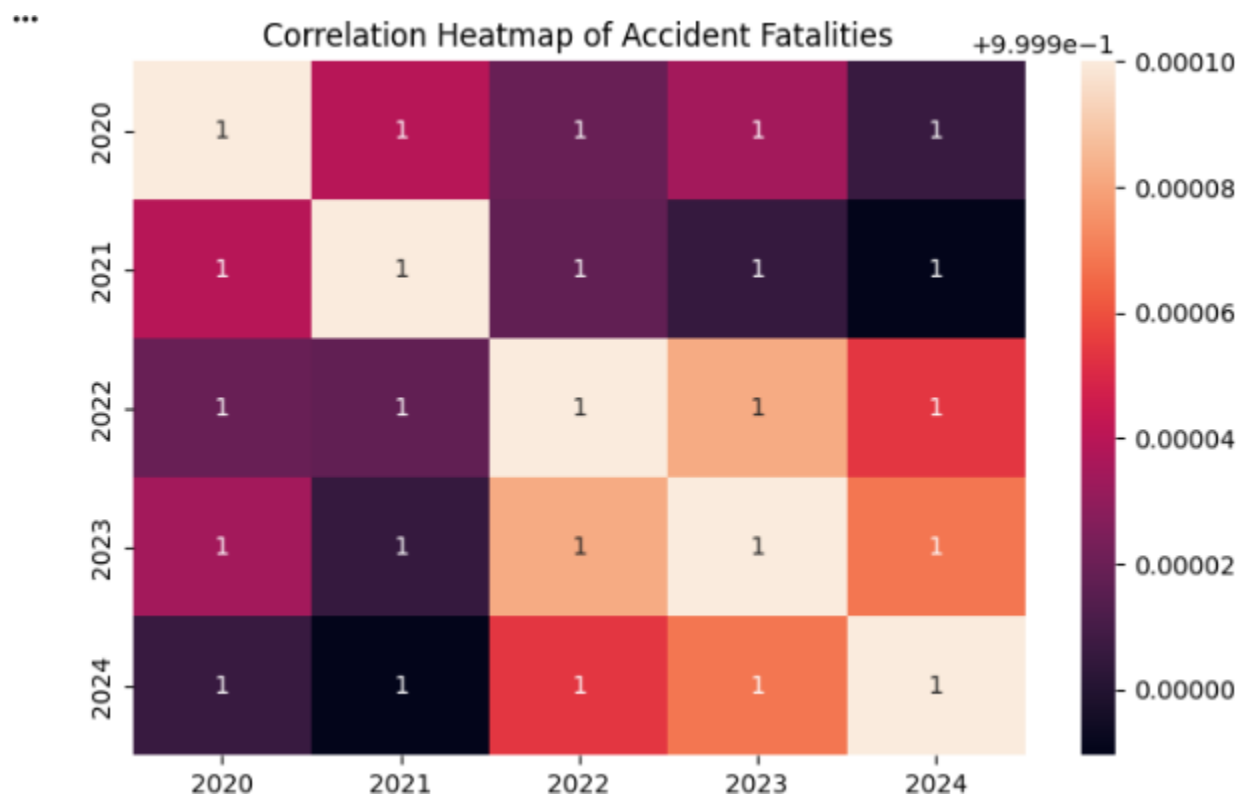
```
▶ # Import required libraries
import seaborn as sns
import matplotlib.pyplot as plt

# Select yearly columns
heatmap_data = df[['2020', '2021', '2022', '2023', '2024']]

# Create heatmap
plt.figure(figsize=(8,5))
sns.heatmap(heatmap_data.corr(), annot=True)

# Add title
plt.title("Correlation Heatmap of Accident Fatalities")

# Show graph
plt.show()
```



Insight-The heatmap shows a strong positive correlation between fatalities across different years. States with high fatalities in one year generally continued to record high fatalities in other years as well.

INSIGHT 7: COMPARISON OF TWO STATES

```
# Import required library
import matplotlib.pyplot as plt

# Compare two states
state1 = df[df['State/UT'] == 'Uttar Pradesh']
state2 = df[df['State/UT'] == 'Tamil Nadu']

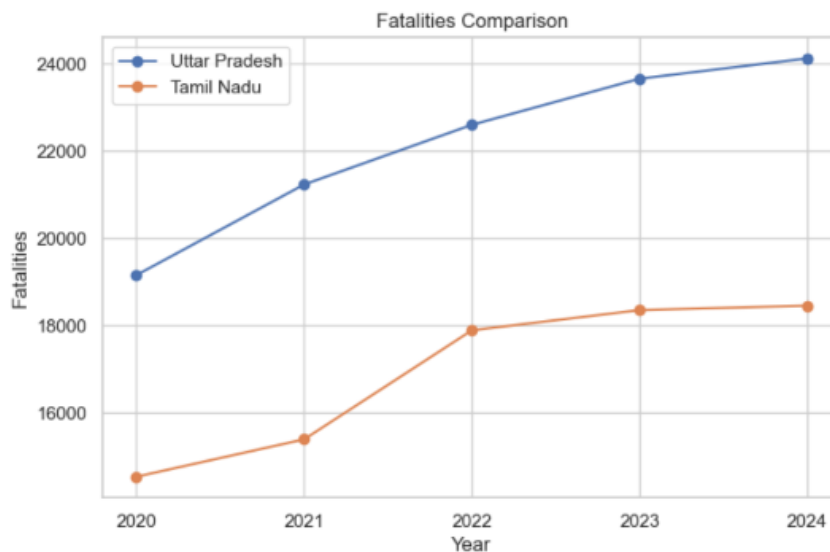
years = ['2020', '2021', '2022', '2023', '2024']

# Plot comparison
plt.figure(figsize=(8,5))
plt.plot(years, state1[years].values.flatten(), marker='o', label='Uttar Pradesh')
plt.plot(years, state2[years].values.flatten(), marker='o', label='Tamil Nadu')

# Labels and title
plt.title("Fatalities Comparison")
plt.xlabel("Year")
plt.ylabel("Fatalities")

# Legend
plt.legend()

# Show graph
plt.show()
```



Insight-The comparison graph shows that both Uttar Pradesh and Tamil Nadu recorded high accident fatalities throughout the years. However, Uttar Pradesh consistently remained higher than Tamil Nadu.

INSIGHT 8: PIE CHART OF ROAD ACCIDENT FATALITIES (2024)

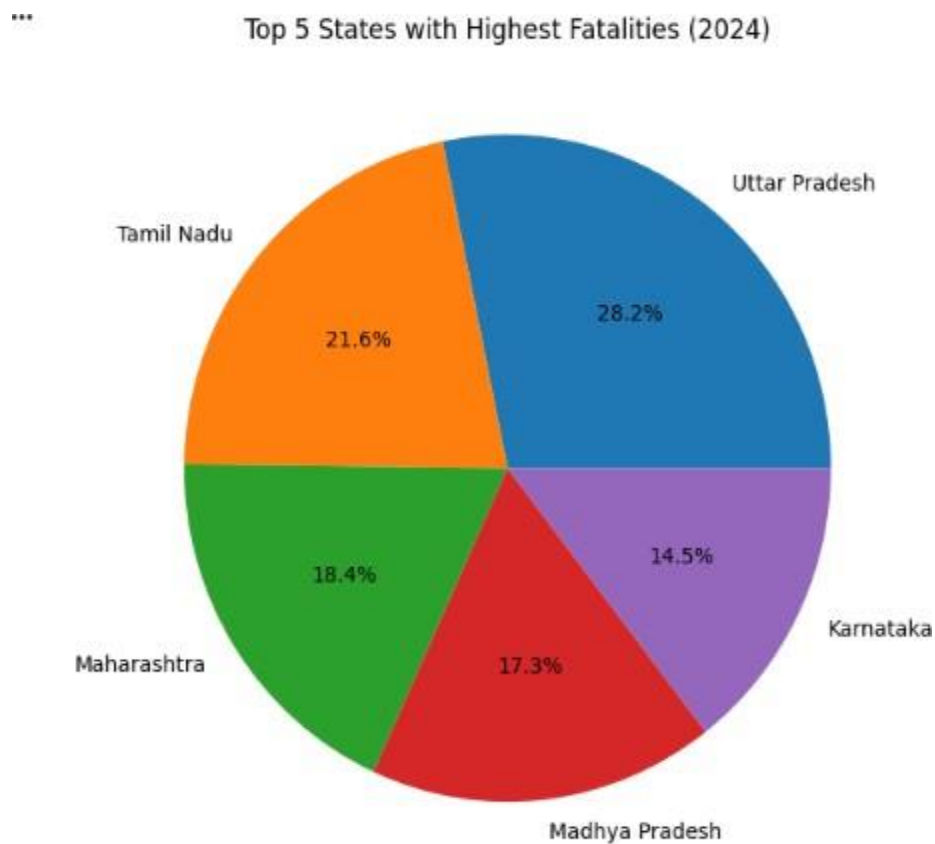
```
▶ # Import required library
import matplotlib.pyplot as plt

# Select top 5 states
top5 = data.nlargest(5, '2024')

# Create pie chart
plt.figure(figsize=(7,7))
plt.pie(top5['2024'], labels=top5['State/UT'], autopct='%1.1f%%')

# Add title
plt.title("Top 5 States with Highest Fatalities (2024)")

# Show chart
plt.show()
```



Insight-The pie chart shows the contribution of the top 5 states to total road accident fatalities in 2024. A large percentage of fatalities is concentrated in a few major states, highlighting the need for stronger road safety measures in those regions.

CONCLUSION

The **Road Accident Fatality Analysis Dashboard** project successfully analyzed and visualized state-wise road accident fatality data in India from 2020 to 2024 using Python and data visualization techniques. The project helped in understanding accident trends, comparing fatalities across different states, and identifying regions with high and low accident rates through interactive charts and graphical representations.

The analysis showed a continuous increase in road accident fatalities over the years, indicating growing road safety concerns in the country. States such as Uttar Pradesh, Tamil Nadu, and Maharashtra recorded the highest number of fatalities, while smaller states and union territories reported comparatively lower accident deaths. These insights highlight the importance of improving traffic management systems, enforcing road safety regulations, and increasing public awareness regarding safe driving practices.

Python libraries such as Pandas, Matplotlib, Seaborn, and Plotly played a significant role in data cleaning, analysis, and visualization. The project also demonstrated the importance of Exploratory Data Analysis (EDA) in understanding patterns and trends before creating dashboards. Overall, the dashboard provides a simple, interactive, and informative way to study road accident fatalities and can support researchers, policymakers, and traffic authorities in making data-driven decisions for improving road safety in India.

REFERENCES

1) [Government of India, Ministry of Road Transport and Highways \(MoRTH\).](#)
Road Accidents in India Reports (2020–2024).

Available at:

[Ministry of Road Transport and Highways](#)

2) [Python Official Documentation](#)

Used for understanding Python programming concepts and syntax.

3) [Pandas Documentation](#)

Used for data cleaning, manipulation, and analysis.

4) [Matplotlib Documentation](#)

Used for creating graphs and visualizations.

5) [Seaborn Documentation](#)

Used for statistical data visualization and advanced plotting.

6) [Plotly Documentation](#)

Used for creating interactive charts and dashboards.

7) [Jupyter Notebook Documentation](#)

Used as the development environment for coding and visualization.

8) [GeeksforGeeks Python Tutorials](#)

Used for learning Python concepts and visualization methods.

9) [W3Schools Python Tutorial](#)

Used for basic Python programming and syntax reference.

10) [DataCamp](#)

Used for learning data visualization, Python programming, and data analysis techniques.